



UNIVERSITY OF GHANA

COLLEGE OF BASIC AND APPLIED SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

DEPARTMENT REGISTRATION APP

STEPHANIE AWURABENA DUNYO (11026494)

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## **INTRODUCTION**

### **1.1 Background of Project**

Student registration is an essential process in higher educational institutions, involving the collection, verification, and management of student data for academic enrollment. Traditionally, these processes have been conducted manually, requiring students to fill out forms, submit documents physically, and move between offices for departmental approval. While the university provides a centralized web portal (MIS Web) for initial course selection, the subsequent departmental clearance and document verification processes often remain manual. These methods are frequently time-consuming, inefficient, and prone to errors such as misplaced physical documents and long physical queues at departmental offices. Although web-based systems exist, they are often not optimized for mobile use, and students still lack a dedicated platform to track their department-level approval status in real-time. With the rapid growth in smartphone usage, mobile applications provide a more convenient and accessible solution for administrative

workflows. This project proposes the development of a Computer Science Department Registration Mobile Application. This app will allow students to synchronize their officially registered courses from the university portal and upload digital "Proof of Registration" documents using their smartphones. The app will enable students to create accounts, fill out departmental forms, upload academic credentials, and track their verification status in real-time. Administrative staff will be able to verify and manage student records and course registration data through a digital interface. This solution aims to bridge the gap between centralized university registration and departmental clearance, improving efficiency, reducing congestion, and enhancing the overall user experience.

## **1.2 Nature of the Project**

This project is a mobile application development project focused on designing and implementing a digital registration system.

The system will consist of:

Mobile Application (Frontend): Developed using React Native to support both Android and iOS platforms.

Backend System: Developed using Node.js (Express) to serve a RESTful API that handles secure authentication, file processing, and verification business logic.

Database: MongoDB will serve as the centralized NoSQL database for storing student profiles and course synchronization metadata, paired with Firebase Storage to securely store uploaded PDF registration slips.

UI/UX Design: Designed using Figma with a mobile-first approach, adhering to a clean 'University Tech' aesthetic matching the institution's visual identity.

The final product will be a fully functional mobile application that enables students to digitally synchronize their registered courses and upload clearance documents, while providing administrators with a secure digital interface to manage, review, and verify student submissions efficiently.

### **1.3 Problem Statement**

Many higher education institutions utilize centralized web portals for academic enrollment, but the department-level clearance, data synchronization, and document verification processes still rely on manual or disjointed workflows. This administrative gap leads to challenges such as long physical queues at departmental offices, excessive paperwork, slow processing times, and difficulty tracking verification records. Students often spend several hours or days completing their departmental registration due to these manual administrative bottlenecks.

Additionally, centralized web-based systems are often poorly optimized for mobile devices, making it difficult for students to easily access and submit department-required records via smartphones. Manual submission of physical slips also increases the risk of data entry errors, misplaced documents, and prolonged delays in verification feedback.

Therefore, there is a need for a mobile-based application that enables students to efficiently synchronize their registered courses and upload digital proof of registration using smartphones, while providing administrative staff with a streamlined digital interface for verification.

### **1.4 General Objective**

To design and develop a mobile-based registration and verification application for the Computer Science Department that enables efficient and convenient synchronization of student data and departmental clearance using smartphones.

### **1.5 Specific Objectives**

The specific objectives of this project are to:

1. Design a user-friendly mobile application interface optimized for smartphone use.
2. Develop a mobile-based student profiling and department registration form.

3. Enable students to upload required documents using their mobile devices.
4. Implement a course synchronization feature that reflects the students officially registered courses.
5. Provide an administrative dashboard for staff to efficiently review, verify, and approve submissions.
6. Implement real-time tracking for registration verification status (e.g., Pending, Approved, Rejected).
7. Integrate push notifications to instantly update students on their clearance status.
8. Store and manage student records and verification data securely in a centralized NoSQL database.

## **1.6 Research Questions**

The project seeks to answer the following research questions:

1. How can departmental registration clearance and course verification processes be improved using a mobile application?
2. How do mobile applications enhance accessibility and convenience for submitting registration proof compared to traditional manual desk checks?
3. What technologies can be used to ensure secure management and cloud storage of student records and uploaded registration data?
4. How can administrative staff efficiently verify and manage student and course registrations digitally?

## **1.7 Justification of the study**

This project provides a practical solution to inefficiencies in traditional departmental clearance workflows. Students will benefit from the ability to submit their registration proof and track their verification status anytime and anywhere without standing in physical queues. Administrative staff will experience an improved, paperless workflow and simplified data management. The system also supports digital transformation in educational institutions by reducing paperwork, improving record accuracy, and increasing operational efficiency at the department level.

## **1.8 Project Stakeholder**

The key stakeholders involved in this project include:

### **Students**

Use the app to create accounts, sync registered courses, upload required slips, and track their clearance status.

### **Administrative Staff**

Use the system to review submitted digital records, verify uploaded registration documents against synced course listings, and approve or reject departmental clearance.

### **University Management**

University management will benefit from improved efficiency in managing student registration processes.

## **1.9 Scope of the study**

The scope of this project focuses on the development of a mobile-based Student Registration Application and Verification for the Computer Science Department. The system will include features such as student account creation, mobile registration forms, course synchronization, registration, document uploads, administrative verification dashboards, and real-time status tracking.

However, the system will not include additional academic management features such as examination results management or financial payment systems.

### **1.10 Assumption & Constraints**

#### **Assumptions**

The system assumes that users will have access to internet-enabled smartphones. It also assumes that students will generate and upload authentic "Proof of Registration" documents directly from the university's official portal.

#### **Constraints**

Possible constraints include limited development time, availability of resources, and the need to ensure system security and data privacy.